Customer retention: RFM segmentation

Team G
Gayatri Jaiswal (Lead)
Ananya
Mahek Motwani
Sneha Chepurwar
Srishty Shree

Newton School of Technology
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Problem statement

Budget spent equally on unequal customers

UK-based online gift retailer · £17.1M revenue · 41 countries · Dec 2009–Dec 2011

The core problem

Retention spend is distributed uniformly across all 5,861 customers — diluting impact on high-value segments and wasting budget on customers unlikely to return.

The objective

Identify which segments drive the most revenue, which face the highest churn risk, and where to reallocate retention spend.

Target: reduce 12-month churn by 10–15% and lift repeat-purchase revenue from the top customer quartile.

Data engineering

01

Source data

UCI Online Retail II dataset · 1,067,371 rows × 8 columns · 2-year transaction window

02

Remove invalid rows

Drop ~270k rows missing CustomerID · filter cancellations (prefix C) · remove non-positive quantities and prices

03

Normalise and dedupe

Exclude non-product items (POST, M, D, BANK CHARGES) · normalise descriptions · deduplicate records

04

Clean output

776,641 valid line-items · 5,861 unique customers · 36,639 invoices

KPI framework

Seven metrics tracked across the RFM analysis — from retention signals to revenue concentration.

KPI	Measures	Signal type	Risk flag
Active Customer Count	Retention pool size	Baseline	Volume drop
Average Order Value	Spend per transaction	Spend depth	Declining basket
Customer Lifetime Value	High-value segments	Value tier	Segment erosion
90-Day Churn Rate	Silent churn rate	Early warning	Undetected churn
Revenue Concentration (Top 20%)	Pareto validation	Concentration	Revenue dependency
Repeat Purchase Rate	Loyalty signal	Loyalty	One-time buyer skew
Country Revenue Mix	Geographic spread	Geo concentration	Market dependency

Five EDA insights

83%

of revenue from top 20% of customers

84%

of revenue from UK, just 91% of customers

Q4

Holiday spike repeated in 2010 and 2011

6× more

International AOV vs. UK (£428 → £2,545)

90 days

Recency cliff: Champions 19 days, At Risk 401

Segmentation methodology

01

RFM scoring

Recency, Frequency and Monetary each scored 1–5, creating a three-dimensional profile per customer.

02

Six segments defined

Champions · Loyal Customers · Big Spenders · Potential Loyalists · At Risk · Lost — six distinct business groups.

03

K-Means clustering

Standardised RFM matrix, elbow + silhouette method → optimal k = 4–5 clusters on unsupervised data.

04

Validated segments

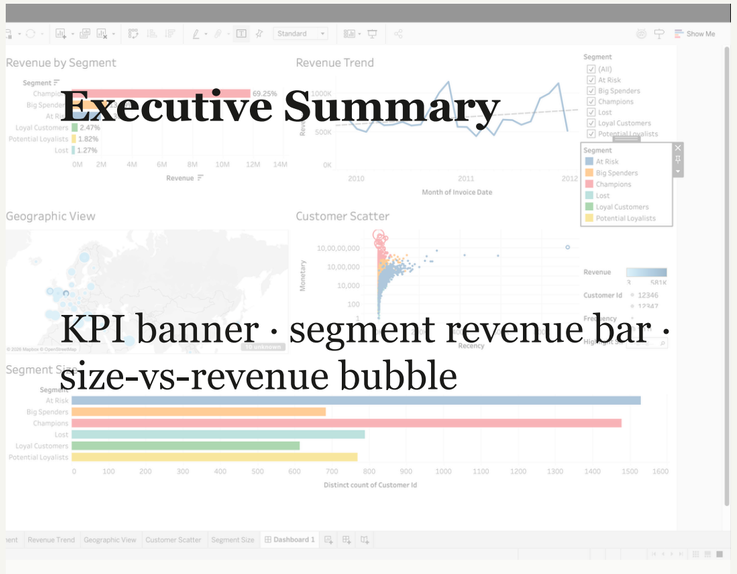
80%+ overlap with rule-based RFM confirms real, separable customer groups — not arbitrary thresholds.

Dashboard overview

Four views, one dashboard

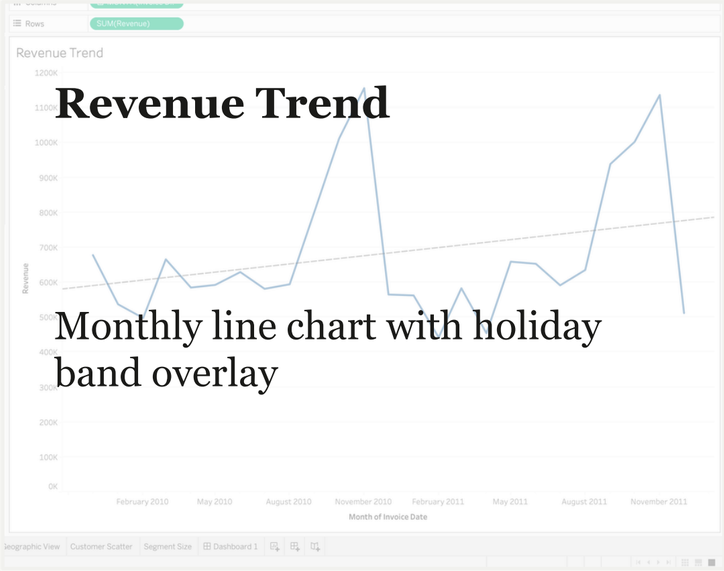
FILTERS

Segment multi-select · Country · Year
Revenue threshold slider



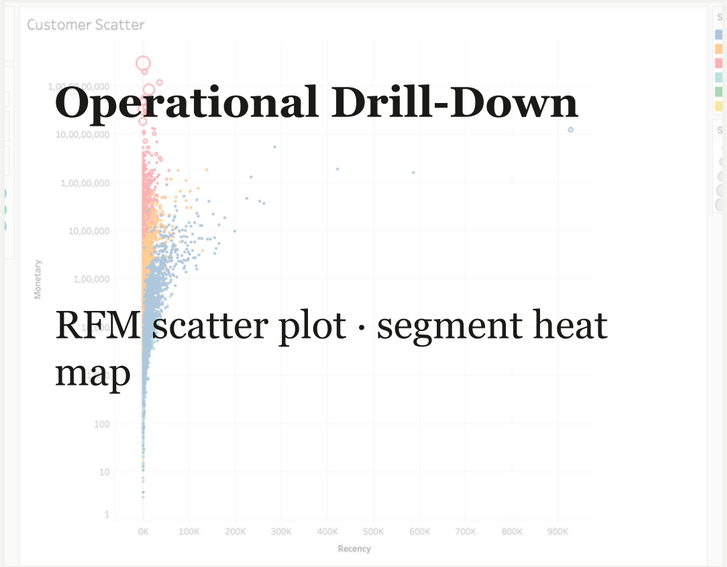
Executive Summary

KPI banner · segment revenue bar · size-vs-revenue bubble



Revenue Trend

Monthly line chart with holiday band overlay



Operational Drill-Down

RFM scatter plot · segment heat map

Geographic

World map with top-10 countries ranked by revenue

Key insights

Top 4 decision-ready insights

RFM segmentation reveals where retention budget creates real return.
Focus on Champions and At Risk — they move the revenue need

Champions

25.2% → 69.2% revenue (£11.8M)

The single highest-leverage retention target.

Low-value tail

Bottom 37% → only 5.6% revenue

Uniform retention spend on this group is unjustified.

At Risk

1,530 customers · £1.93M dormant

Largest segment (26.1%). Biggest win-back opportunity.

International edge **4–6× more revenue per customer** International customers earn more revenue per customer than UK. Premium positioning works — expansion opportunity.

Five recommendations

REC 01

Champion Protection Program

Premium loyalty tier, dedicated CSM, alert at 30 days inactive

Retain

REC 02

At Risk win-back

Personalised reactivation + 15% discount sequence

Revive

REC 03

Reallocate from Lost

Remove low-value segments from active campaigns

Cut

REC 04

International premium expansion

Targeted ads in Netherlands, Australia, Germany, France

Grow

REC 05

Second-purchase upsell

60-day post-purchase nudge, push frequency up

Upsell

£1.35M–£1.45M

projected Year-1 net impact

+£485k–£530k

Champion protection

+£530k–£550k

Upsell campaigns

+£260k–£275k

At-risk win-back

+£20k–£30k

Reallocate retention budget

+£50k–£70k

International expansion

Spend: £90k–£185k (<1% of revenue) · ~8–9% of annual revenue defended or recovered · Priority: Recs 1, 2, 5 (fastest payback); Recs 3, 4 (quick to launch)

Limitations and caveats

No demographic data

Segments are behavioural only — age, gender, and income are unknown.

No acquisition channel info

CAC and channel ROI cannot be measured or attributed.

Two-year window

The limited timeframe restricts cohort depth and survival analysis.

Single retailer, single category

Findings are context-specific and may not generalise to other retail settings.

GBP only

No currency normalisation — cross-market spend comparisons are not valid.

Next steps

Four analyses to sharpen what we've built

Predictive churn model

Flag customers before they hit At Risk — shift retention from reactive to predictive.

Cohort survival analysis

Map systemic churn timing to identify when customers are most at risk of dropping off.

A/B test infrastructure

Validate the five recommendations with controlled tests before full-scale rollout.

Real-time dashboard

Make segment movement visible daily — not historically — so the team can act in the moment.

Five recommendations · £1.35M–£1.45M projected annual impact · under 1% revenue marketing spend · all launchable within 4–8 weeks.